Erfan **Khadem**

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An EE student, troubleshooting oscillators that don't and amplifiers that do

Education

Bachelors of Science in Electrical Engineering — *2022 - 2026*
Sharif University of Technolog — *Tehran, Iran*

- GPA: 18.69 / 20, Top 10%
- GPA of EE Courses: 19.15 / 20
- Elective Courses: DSP | Compressed Sensing | MCMC Methods & Graphical Models | Operating Systems Lab | Cryptography

Highschool Diploma in Physics and Mathematics — *2019 - 2022*
Dastgheib 1 Highschool of Shiraz — *Shiraz, Iran*

- GPA: 19.75 / 20

Projects & Publications

Non-Uniform R-2R DAC Design — *2025*
Publication Pending Review — *IEEE TIM*

- A novel, tolerance resilient, fast and low-noise precision DAC design built using non-precision off-the-shelf components
- A sample circuit was built and measured to have `15.4 ENOB` with artificially induced 1% tolerance in resistors
- Consistently achieves `< 1.3 ppm` max measured error in 2 hour windows and `< 2.8 ppm` in 12 hour windows
- Has a settling time of under $10\mu s$ to $100\mu V$ and a maximum glitch impulse of under 3.5 nV-sec
- Fast self calibration without any external references or ADCs for temperature drift compensation
- Significantly eases the requirements on the TIA offset and the analog switches of the classic R-2R implementation

Precision Low Temperature Coefficient and Low Noise Voltage Reference — *2025*
Used in the Non-Uniform DAC Design

- Using an LM199 as the reference with additional filtering
- Achieves $10\mu V$ peak-to-peak noise in 24 hour windows
- Achieves an end-to-end temperature coefficient of under 0.3 ppm/K
- Less than $2.2\mu V$ RMS broadband noise from 3 Hz to 300 kHz

Low-Glitch Precision Sample and Hold Circuit — *2025*
Used in the Non-Uniform DAC Design

- At least an order-of-magnitude better than `LF398` in most metrics
- A hold-step of under $100uV$ in the entire output voltage range of $-5V$ to $10V$ and over the full temperature range
- Full power bandwidth of more than 400 kHz, Rise-time and Fall-time of under $1.2\mu s$ for an $8V$ step
- Droop rate of less than $1.1mV/\text{sec}$ over the full voltage range, corresponding to a total leakage current of under $11pA$
- Droop rate of less than $120\mu V/\text{sec}$ near $0V$ hold value
- Small hold moment and sample moment glitch-impulses, smaller than 1.5 nV-sec without internal or external filtering

PRISM: A new scheduling method for single-hop network discovery — *2025*
Pending Submission

- A deterministic neighbor discovery framework for single-hop wireless networks
- Achieves $O(L(1+\delta)\log K)$ discovery time with High Probability, $O(L^2 \log K)$ deterministic performance
- Significantly outperforms CSMA and the ALOHA protocol

RP2040 based Intel 8080 emulator — *2024*
erfan-khadem/rpi8080 — *Github*

- Implemented a full-fledged Intel 8080 emulator on the Raspberry Pi Foundation's RP2040.
- The project has been used since Spring 2024 in the computer architecture laboratory for educational purposes.
- On-chip debugger, virtual input and output ports and accompanying PC software for ease of use.

Breathing Rate Detection Using WiFi Channel State Information

2024

A Hobby Project

- Breathing rate detection using a common off the shelf intel network card.
- Achieves an accuracy of more than 98 % even in highly congested areas.
- Runs transparently and does not disrupt the normal user activity.
- Uses ICA for Blind Source Separation

Linear OptoCoupler

2024

Electronics Lab Project

- A highly linear, optically isolated analog voltage isolator using `HCNR201`.
- Achieves a maximum signal bandwidth of more than $1\ \mathrm{MHz}$.
- Less than $-70\ \mathrm{dB}$ THD at $1\ \mathrm{kHz}$ and full swing.
- PCB was designed using KiCAD.

SSH Anti-Censorship Tool

2023

SSH-liberty-bridge — *Github*

- Implemented in go with the user management component written in python.
- Has been used successfully for bypassing GFW restrictions in Iran and bringing internet freedom to the citizens.
- Integrated in Hiddify's excellent GFW bypass solution.

Advanced Algorithms in Rust

2022

TheAlgorithms/Rust — *Github*

- Implemented advanced graph algorithms such as centroid decomposition and max-flow.
- Implemented various cryptographic algorithms including Salsa20 and HMAC.
- Improved and optimized the implementation of various algorithms like SHA256 and DSU.

Professional Experience

Embedded Hardware and Software Developer

2025-2026

BIOSEN Group — *Tehran, Iran*

- Developed and worked on a Gel Doc medical imaging system used for visualization and characterization of proteins and nucleic acids (RNA / DNA). This involved optimizing the firmware on the hardware side and writing a cross-platform user interface with image processing capabilities.
- Technologies:
  - Embedded C / C++
  - Video Hardware Encoding / Decoding
  - OpenCV
  - Realtime PC-MicroController Communication

Embedded Hardware and Backend Developer

2023

Home Automation Project at National Elites Foundation — *Tehran, Iran*

- Worked alongside a team of computer security experts, electrical engineers and developers to build home automation appliances and smart gateways using TI's Zigbee chips and Espressif System's ESP32 and ESP8266 microcontroller series
- Technologies:
  - FreeRTOS
  - Zigbee
  - Websocket
  - Embedded Linux

Fullstack Developer and Systems Administrator

2022 - 2023

Abarkelas — *Tehran, Iran*

- Fix bugs and improve UX at our frontend using SSR and Vue.js
- Improve monitoring and alerting for our backend systems
- Implement self-healing procedures for our backend systems
- A/B test new features to guide our product development process
- Technologies:
  - Django REST Framework
  - Vue.js
  - Docker
  - Postgres + Redis
  - Grafana + Prometheus

Certificates

2021 **Scientific Computing with Python**, Free Code Camp *view*
2021 **Javascript Algorithms and Data Structures**, Free Code Camp *view*

Skills and Topics of Interest

Software Stack C/C++ (Embedded/FreeRTOS/Linux) | Scientific Python | Svelte/Vue.js | Go

Mathematical Interests Compressed Sensing | Stochastic Processes | Monte Carlo Methods | DSP

CAD Software KiCAD | Hyperlynx (Power Integrity) | Qspice | MicroCap 12

Embedded Systems Raspberry Pi 5 | STM32 {F4, H7, G4, WB55} | ESP32 | RP2040 / RP2350

Languages Persian (Native) | English (TOEFL 106/120 — L:29, R:30, W:25, S:22)

Hobbies Chess | Bicycle | Electronics | Mathematics